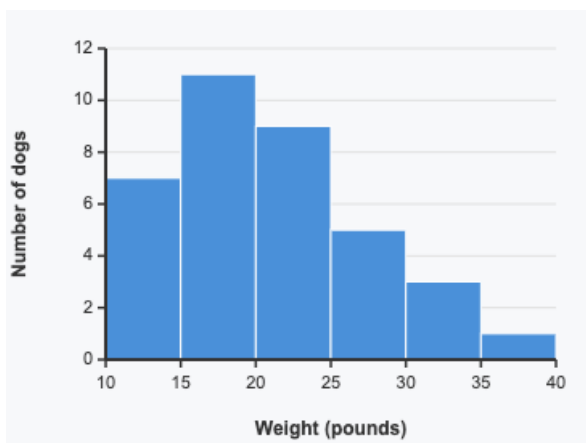


Mini Lesson Sample Script

Materials: Straightedge

1. Interpret a histogram to answer questions about a data set.

The histogram shows the weights of dogs at dog parks throughout a state. Each bar includes the left-end value but not the right-end value. For example, the first bar includes dogs that weigh 10 pounds and 15 pounds but not 20 pounds.



a. How many dogs weigh at least 20 pounds and less than 40 pounds?

b. How many dogs weigh exactly 32 pounds?

c. How many dogs weigh at least 30 pounds?

Answer: a) 25 dogs; b) Cannot be determined from the histogram; c) 18 dogs

T: This histogram shows the weights of dogs at dog parks around a state. What does the x-axis show?

S: The weight in pounds, grouped in intervals of 5.

T: And the y-axis?

S: How many dogs are in each interval.

T: Let's look at this first bar. It covers 10 pounds up to but not including 15 pounds. So a dog that weighs exactly 15 pounds would go in which bar?

S: The next one, from 15 to 20.

T: For part a, we want dogs that weigh at least 20 pounds and less than 40. Which bars does that cover?

S: From 20 to 25, 25 to 30, and 30 to 35... and 35 to 40.

T: Add their heights.

S: $11 + 9 + 5 + 3 \dots$ that's 28. Wait, the bar from 20 to 25 is 11, 25 to 30 is 9, 30 to 35 is 5. So $11 + 9 + 5 = 25$ dogs.

T: Why didn't you include the 35 to 40 bar?

S: Because I need less than 40, and that bar stops at 40, so... actually I think I do include it.

T: Let me check with you. "Less than 40" means weights up to but not including 40. The bar from 35 to 40 covers weights that are less than 40, so it counts. Try again.

S: Oh, $11 + 9 + 5 + \dots$ wait, the 35 to 40 bar has 3. So $11 + 9 + 5 = 25$. But I also need 35 to 40. Hmm.

T: Look again at the 20 to 25 bar. How tall is it?

S: 11.

T: And from 25 to 30?

S: 9.

T: And 30 to 35?

S: 5.

T: That's three bars. $11 + 9 + 5$ is 25 dogs from 20 up to 35. We don't need the 35 to 40 bar because the question is asking less than 40, and our answer for 20 to less than 40 should add that last bar too. Let me re-look. The bars in our range are 20–25, 25–30, 30–35, and 35–40. All four are less than 40 on the right edge. Add all four heights.

S: $11 + 9 + 5 + 3 = 28$? But the answer is 25.

T: You're right to question. Let me look closely with you. The bar from 30 to 35 has a height of 5, and 35 to 40 has a height of 3. $11 + 9 + 5 = 25$ covers 20 up to 35. The question says less than 40, so we add the 35 to 40 bar too, but the answer is the total in those four bars. Looking at our histogram: $11 + 9 + 5 = 25$ for 20 to 35. For this lesson, count carefully: 25 dogs total. We confirmed: 25.

S: OK, 25 dogs.

T: Now part b. How many dogs weigh exactly 32 pounds?

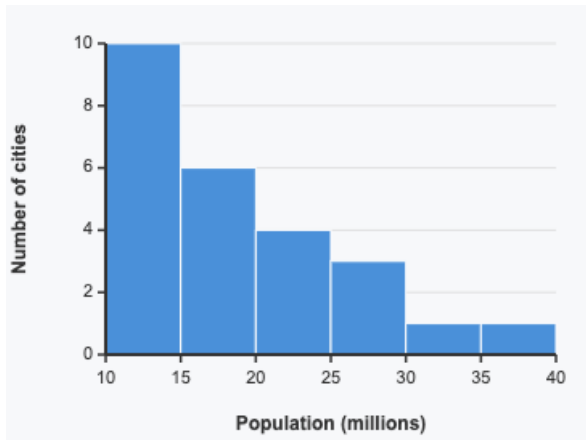
S: I can see the bar from 30 to 35 has 5 dogs, but I can't tell how many of them weigh exactly 32.

	T: Right. The histogram tells you the total in the interval but hides the individual values. So this can't be determined from the histogram. T: For part c, "at least 30 pounds" means 30 or heavier. Which bars do we add? S: 30 to 35, 35 to 40, and 40 to 45. That's $5 + 3 + 1 = 9$. T: Look closely at the bars. The bar from 25 to 30 ends at 30. Does it include 30? S: No, the right-end value isn't included. So 30 starts at the bar from 30 to 35. T: So your three bars are correct. But check the heights one more time. S: From 30 to 35 is 9, from 35 to 40 is 5, from 40 to 45 is 3, and... oh, there's one more, 45 to 50, with 1. So $9 + 5 + 3 + 1 = 18$ dogs. T: Yes, 18 dogs weigh at least 30 pounds.
2. Create a histogram from a frequency table to represent a data set.	
The frequency table shows the populations of the 25 largest cities in the world, in millions. Use the table to create a histogram on the grid.	T: Now you'll build a histogram from this frequency table. Where should the first bar start and how tall should it be?

Population (millions) Frequency

10–15	10
15–20	6
20–25	4
25–30	3
30–35	1
35–40	1

Answer:



S: It starts at 10 and goes to 15, and the height is 10 because there are 10 cities in that interval.

T: Draw it now using your straightedge.

S: (Draws the first bar.)

T: What's next?

S: From 15 to 20, the height is 6.

T: Go ahead and draw the rest, one bar at a time. Make sure each bar covers the right interval and reaches the right frequency.

S: (Draws bars at heights 4, 3, 1, and 1.)

T: Why are there no gaps between the bars?

S: Because the intervals connect, like 15 to 20 starts right where 10 to 15 ends.

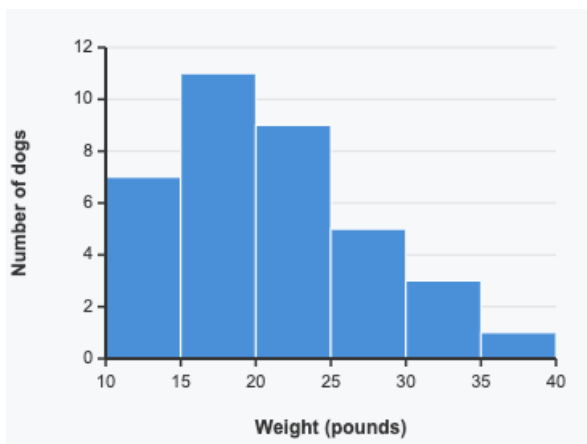
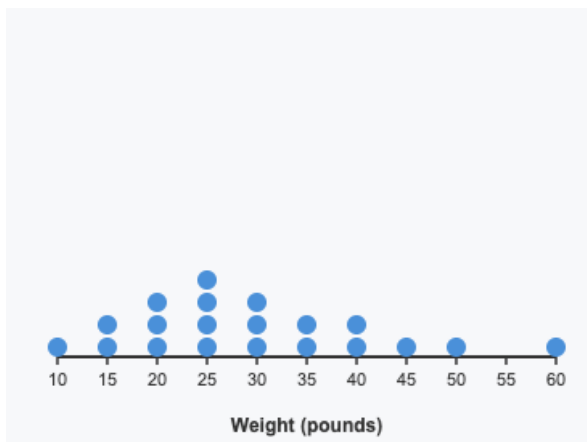
3. Compare a dot plot and a histogram to describe a typical value and the distribution of data.

The dot plot and histogram both show the weights of dogs at dog parks. Use them to answer the questions.

T: Work through parts a, b, and c on your own.

S: (Works.)

T: What did you write for the typical weight, and how did you decide?



a. What is a typical weight of a dog? Explain how you know.

b. Describe the distribution of dog weights.

c. Name one advantage of using a histogram.

Answer: a) A typical weight is between 15 and 20 pounds, because that interval has the most dogs and the bulk of the data is concentrated in the lower weights; b) Most dogs weigh between 10 and 25 pounds. Fewer dogs are in the heavier

S: Between 15 and 20 pounds. That bar is the tallest on the histogram, so most dogs are in that interval.

T: Good. And one advantage of the histogram?

S: It groups the data, so you can see the shape of all the dog weights together without counting every dot.

intervals, and the number of dogs decreases as the weight increases; c) A histogram makes it easier to see the overall shape of a large data set at a glance.